

Mutualism

Learning objectives:

1. Understand the costs and benefits of different mutualistic relationships
2. Quantify positive interactions with consumer-resource model

density dependence, other plant species, and natural enemies limit population growth

today focusing on processes that would increase population growth

Facilitation

- Interspecific, nurse plants in arid environment

Draw number of neighbors on x, individual fecundity on y, multiple lines for multiple stress levels (rain shelter, natural, supplemental rain): stress increasing, favorable decreasing; slopes highlight this change

- Intraspecific, plant-soil feedbacks

figure out plot for this

Animal-mediated pollination

- pollinators get nutrients, plants get fertilization
- Benefits: genetic variation *outcrossing, compared to self-pollination, wind pollination*
- Costs: dependent on animal density, cross-contamination *pollen from other plants not your own*

figure out plot for this

Mycorrhizal fungi

- Endomycorrhiza: fungi in root cells, arbuscular mycorrhiza *found in many plant taxa, others specialize*
- Ectomycorrhiza: fungi influence root cells but not inside cell, trees and shrubs
- fungi get carbohydrates, plants get nutrients
- Benefits: drought tolerance, salinity tolerance, growth, increased innate immunity
- Costs: increased carbon costs *fungi are carbon sink*

Modeling

- v - visitation rate
- h - handling time per flower
- B - pollinator abundance
- M - mutualism benefit, proportion

$$M = \frac{vB}{1 + vhB}$$

- effect on fecundity

- fecundity increased if pollinators present

$$f_{eff} = f_{\mu}(1 + M)$$

from lab (assumes herbivore is pollinator??):

$$M = \frac{pH}{1 + ph_p H}$$